Screenplay Intelligence & Promo Finder for Vertical Microdramas

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Prototype Web App: https://kuku-promo-finder-saurabh105.streamlit.app/

Source Repository: https://github.com/100rabhsolanki/kuku-promo-finder | Dataset: 50 Episodes (3,727 blocks)

1. PROBLEM CHOICE & EDITORIAL CONTEXT

Why Problem 1: In vertical microdrama publishing, acquisition ads on TikTok, Reels, and Shorts drive user growth, but creative discovery is a manual bottleneck. Video editors currently spend 1.5 to 2 hours per show scrubbing through 50 episodes to spot hooks, reveals, and cliffhangers. I built this tool as an editor-assist scout to cut this discovery phase from 100 minutes to under 5 minutes. The tool parses planned screenplay timestamps, extracts 20-60s candidate moments starting with immediate verbal/physical shock and cutting at unresolved cliffhangers, and presents editors with ranked cards including exact timestamps, evidence excerpts, and quality warnings.

2. CORE OPERATIONAL ASSUMPTIONS

Screenplay vs. Final Video Runtime: Screenplay timestamps (Start: MM:SS.SS | End: MM:SS.SS) represent planned pacing during writing. Actor paus scene transitions, and bumpers introduce minor drift. The tool treats script timestamps as reliable anchors for beats, leaving final 0.5s trims to the editor.
Scroll-Stopping Hook Window: On vertical social feeds, viewers drop off within 2.5 to 3.0s. A promo whose confrontation or reveal occurs after 3.5s fails as an ad, regardless of how strong the rest of the scene is.
Single Continuous Clips vs. Multi-Cuts: Single continuous scenes provide optimal narrative clarity for cold scrollers. I prioritized continuous stretches while supporting curated cross-episode pairs for high-contrast storylines (e.g., Altar Betrayal in Ep 1 paired with Underdog Marriage in Ep 4).

3. ARCHITECTURE: WHAT I KEPT AS CODE VS. WHAT I GAVE TO AI

I deliberately avoided dumping entire screenplays into an LLM. Large models hallucinate timestamps, are slow, and struggle with window math:

DETERMINISTIC CODE & RULES	SCORING HEURISTICS & AI MODEL
- Regex parsing of 3,727 camera, dialogue, and action blocks across 50 episodes - Strict 20-60 second windowing math and multi-track concurrency overlap detection - Non-Maximum Suppression (NMS, 0.35 IoU) eliminating 6,154 duplicate variations	- Hook immediacy scoring: linguistic shock, verbal insults, and sound stingers in 0-3.5s - Cliffhanger urgency scoring: cut-to-black peaks, unanswered threats, and reveal cuts - Standalone clarity classification and jargon penalties (demoting deep-lore exposition)

4. MATHEMATICAL SCORING MODEL & RANKING RUBRIC

Composite Score = 0.30(Hook) + 0.25(Cliffhanger) + 0.20(Standalone Clarity) + 0.15(Intensity) + 0.10(Context)

1. Hook Immediacy (0-10): Scored by linguistic shock (e.g. 'You're the Duke?', 'Married a beggar!'), physical aggression (slaps, blades), and audio hits within the first 3.5s. Decays by 40%/s for delays beyond 3.0s.
2. Cliffhanger Urgency (0-10): Unanswered questions, cut-to-black scene peaks, interrupted revelations, and unresolved physical challenges.
3. Standalone Clarity (0-10): Rewards universal archetypes (wedding humiliation, undercover Duke); penalizes niche medical/lore jargon.
4. Temporal Deduplication: Retains top-scoring moments while eliminating 6,154 overlapping time-shifted candidate windows via NMS.

5. TOP-RANKED PROMO CANDIDATES (FROM FULL 50-EPISODE RUN)

Rank 1: Episode 45 (00:19.40-01:14.00 | 54.6s | Score: 9.02/10 | High Clarity): Undercover Duke Reveal.
Hook (00:22.90): JULIANA: 'You're the Duke?' | Cliffhanger (01:10.50): CATHERINE: 'Wife?!' as Alexander severs her chains and claims head of house.
Why it works: Quintessential microdrama reveal. High emotional payoff that stops scrollers instantly and cuts right before the antagonist can react.
Rank 2: Episode 42 (00:14.60-01:12.40 | 57.8s | Score: 8.84/10 | Medium Clarity): Shattered Cathedral Gates.
Hook (00:18.00): NOBLEMAN: 'Guards! Arrest this imposter!' | Cliffhanger (01:12.40): MOTHER: 'Let everyone hear me! This is my son Alexander, the Duke!'
Why it works: Public status reversal with high dramatic escalation and immediate visual/verbal stakes.
Rank 3: Episode 11 (00:26.90-01:24.00 | 57.1s | Score: 8.15/10 | Context-Dependent): Class Betrayal Confrontation.
Hook (00:29.30): GERALD: 'You married a beggar!' | Cliffhanger (01:24.00): GERALD: 'Going to strike a gentleman in front of his wife?' before fistfight.
Why it works: Visceral class bullying that provokes audience outrage and cuts right before physical retaliation.

6. WHERE THE TOOL PICKS BADLY (REAL FAILURE MODES & EDGE CASES)

Automated systems fail when narrative context cannot be deduced from isolated words. Auditing the 144 picks revealed 4 real failure modes:
Failure 1 (Medical Caretaking False Positive, Ep 8 & 9): Words like 'blood', 'wound', and 'fever' scored 8.8/10 on intensity, but Juliana was quietly brewing herbal tea and cleaning Alexander's leg. It has zero hook energy for TikTok. Fix: Negative weights for non-combative medical scenes.
Failure 2 (Lore-Trapped Climax, Ep 29): High dungeon tension scored well, but dialogue revolved around 'charcoal antidotes' and 'apothecary ledgers' A cold TikTok scroller has no idea who Rupert is. Fix: High-context jargon penalty; candidate flagged as 'Standalone Clarity: Low' for editor text overlay.
Failure 3 (Establishing Shot Pacing Lag, Ep 7): Insult occurs at 00:08.50 after silent door opening. If clipped from 00:00.00, scrollers bounce. Fix: Window start dynamically anchors directly to the exact timestamp of the hook beat rather than blindly using the scene heading start.
Failure 4 (Mid-Sentence Dialogue Clipping): Blindly cutting at exactly 60.0s sliced dialogue mid-syllable. Fix: All cuts snap to completed dialogue bloc

7. HUMAN-IN-THE-LOOP EDITORIAL PROTOCOL

The tool acts as an algorithmic scout; the human video editor confirms final cuts via a 3-step checklist:

1. Visual Jolt Audit: Verify opening 0-2s has immediate motion or punch-in; front-trim 0.5s if actor hesitation exists.
2. Audio Tail Audit: Confirm dialogue does not cut mid-sentence; verify music swell resolves into an unresolved blackout cut.
3. Channel Routing: Route high-betrayal/slap clips (Ep 1, 11) to TikTok; route royal reveal/action (Ep 42, 45) to YouTube Shorts.

8. AI TOOLS USED & WHERE THE MODEL GOT SOMETHING WRONG

I used Gemini 3.8 Flash and Claude 3.5 Sonnet to help analyze microdrama archetypes and calibrate keyword weights.
Where the AI Got It Wrong: When analyzing Ep 1, an LLM flagged Juliana clenching her bridal gown at 02:11.30 as a 'resolved emotional ending' because Gerald had finished his monologue and walked away. The AI mistook the physical end of a speech for emotional closure, missing that Juliana's silent, hardening glare into a cut-to-black drone swell is the quintessential cliffhanger setup. I fixed this with an explicit heuristic rule.

9. PRODUCTION SCALING & ECONOMICS

Execution Speed: The Python engine processed all 50 episodes (102.6 minutes of drama, 3,727 blocks) in 0.91 seconds.
Cost at Scale: $0.00 for rule-based parsing and NMS deduplication. Connecting an LLM API (like Gemini Flash) for social ad copy generation on the top 15 candidate cards costs less than $0.01 per show, saving hours of manual labor.